Anteneh Getachew Yitayal

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Education

University of Trento — Trento, Italy
MSc in Data Science — *2024 – Present*

Addis Ababa Science and Technology University — Addis Ababa, Ethiopia
BSc Electrical Engineering (Computer Engineering Focus) — *2016 – 2021*

Experience

Software Developer — Jan 2023 – Aug 2024
Synapse Software Solutions — *Addis Ababa, Ethiopia*

- Developed the Electronic Public-Private Dialogue (EPPD) portal for Ethiopia's Ministry of Trade & Regional Integration, funded by GIZ and the EU; platform serves federal, regional, and city government bodies
- Built full-stack web systems (PHP Laravel, MySQL, JavaScript, Vue.js) for government clients, including the Ethiopian Wildlife Conservation Authority

Machine Learning Research Intern — Mar 2021 – Aug 2021
Chapa Financial Technologies — *Addis Ababa, Ethiopia*

- Researched NPL prediction using Random Forest, Decision Tree, SVM, and XGBoost on real banking datasets
- Analysed feature importance for credit risk modelling; research presented at NeurIPS 2021 Workshop and PanAfriconAI 2022

Certifications & Training

Google Developer Machine Learning Bootcamp – Sub-Saharan Africa — Aug 2022 – Jan 2023
Google & Gebeya — *Competitive Selection Program*

- Selected for a competitive Google-sponsored ML bootcamp; completed the Deep Learning Specialization and TensorFlow Developer Specialization (DeepLearning.AI, Coursera) fully funded by Google
- Received mentorship from Google ML researchers and professionals from YouTube and LinkedIn; applied skills through capstone projects

Stanford Machine Learning Specialization: Andrew Ng, Coursera (2022) — Self-initiated

Projects

Image Retrieval with SimCLR & Supervised Contrastive Learning | *PyTorch, ResNet-50, FaceNet, EfficientNet*

- Implemented and compared image retrieval pipelines using SimCLR-pretrained ResNet-50, FaceNet, and EfficientNet
- Fine-tuned models with Supervised Contrastive Loss (SupCon), achieving 97.77% top-3 accuracy on Intel Image dataset
- Analyzed embedding spaces with t-SNE; investigated the effect of domain alignment and data scarcity on contrastive learning

Bayesian Active Learning with MC Dropout | *PyTorch, CNN, CIFAR-10*

- Built a CNN with Monte Carlo Dropout for uncertainty-based active learning on CIFAR-10
- Implemented and compared 6 query strategies: Random, Entropy, Margin, BALD, Confidence, and Consensus Entropy
- Consensus Entropy achieved the best test accuracy (72.56%), demonstrating the value of Bayesian uncertainty estimates

Parallel Conjugate Gradient Solver (MPI) | *C, MPI, HPC*

- Implemented Jacobi-preconditioned CG solver for large sparse SPD systems using MPI z-slab domain decomposition
- Built a pipelined variant with `MPI_Iallreduce` and halo exchange, reducing communication from $O(N^3)$ to $O(N^2)$; benchmarked up to 96 MPI ranks on HPC cluster

BiRank & Matrix Factorization Movie Recommendation | *Python, Pandas, SciPy, NumPy*

- Implemented and compared BiRank and Matrix Factorization on MovieLens 100K; BiRank achieved 57% HR@10 and 17% NDCG@10 vs 48% and 7.2% for MF; MF scaled to 1M where BiRank failed due to memory constraints, revealing a ranking quality vs scalability trade-off

NPL Prediction and Feature Importance Analysis | *Python, Scikit-learn, XGBoost*

- Built classification models (Random Forest, SVM, XGBoost) for non-performing loan prediction; analysed feature importance for financial risk assessment; published on arXiv

Technical Skills

Languages: Python, PHP, SQL, JavaScript, C/C++
Machine Learning: Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, Matplotlib
Deep Learning: CNNs, Contrastive Learning, Active Learning
Data & Big Data: Apache Spark, Kafka, Flink, Airflow, Docker, Git, Linux
HPC & Systems: MPI, OpenMP, C (parallel computing)
Web Development: Laravel, JavaScript, FastAPI, Django, Bootstrap, Vue.js